Construction Chemicals

SUBMITTAL DOCUMENT

Addmix VAE

Vinyl Acetate–Ethylene (VAE)
Polymer Dispersion



Flexible	Rapid Setting	Adhesion	MFFT
Excellent elongation	Fast bonding	Plastics & films	0 °C

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

Construction Chemicals

Contents

1. Method Statement 3

2. Technical Data Sheet 8

3. Test Reports 11

4. Approvals & Certificates 12

METHOD STATEMENT

Addmix VAE — Vinyl Acetate–Ethylene (VAE) Polymer Dispersion

Project Name		**Document No.**	OMX-MS-ADDMIXVAE
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Addmix VAE	**Status**	For Approval

1. Purpose

This method statement defines the procedure, resources, quality checks and safety measures for the receipt, storage, handling, use and quality control of Addmix VAE in the production of water-based adhesives.

2. Scope of Work

This method statement covers the receipt, storage, handling, formulation use and quality control of Addmix VAE polymer dispersion in adhesive production.

- Receipt and inspection of the material.
- Storage and handling in the plant.
- Laboratory trials to set the formulation.
- Use in production batches.
- Quality control of finished adhesives and documentation.

3. Definitions

Term	Meaning
Dispersion	Polymer particles dispersed in water.
MFFT	Minimum temperature at which the dispersion forms a continuous film.
Open time	Time after application during which bonding is possible.
Formulation	Approved recipe of the adhesive.

4. Roles & Responsibilities

Role	Responsibility
Production Manager	Overall responsibility for planning and compliance with approved formulations.
Laboratory / R&D	Develops and approves formulations and tests.
Production Supervisor	Supervises weighing, mixing and batch records.
QC Inspector	Tests incoming material and finished batches.

Role	Responsibility
HSE Officer	Checks PPE, safe handling of drums and IBCs and spill control.
Storekeeper	Receives and stores material under the stated conditions; controls batch numbers.

5. Materials

Material	Description
Addmix VAE	VAE dispersion, 200 / 220 kg drums, IBCs

5.1 Key product data

Property	Typical Value
Polymer	Vinyl acetate–ethylene copolymer
Protective colloid	Polyvinyl alcohol (PVOH)
Appearance	Milky white dispersion
Solids content	54 – 56 %
Viscosity (25 °C)	1300 – 2000 mPa·s
pH	4 – 6
Density (20 °C)	approx. 1.07 g/cm³
Minimum film-forming temperature	0 °C

6. Tools & Equipment

- Drum pump or IBC valve
- Calibrated scale
- Low-speed mixer
- Viscometer
- PH meter
- Laboratory equipment
- Spill kit and absorbent material
- Sample containers for retained samples

6.1 Personal protective equipment

- Gloves
- Goggles
- Overalls
- Safety shoes

7. Storage & Handling

- 9 months in original, unopened packaging. Store at +5 °C to +30 °C; protect from frost and direct sunlight.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Keep containers tightly closed and protect from frost and direct sunlight.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Formulation approved by the laboratory.
- Mixing equipment clean and calibrated.
- Scale calibrated.
- Batch records prepared.
- Production team briefed on the procedure and safety.

9. Laboratory Trial

1. Prepare laboratory batches of the adhesive formulation.
2. Measure viscosity, pH and solids.
3. Test open time, setting speed, bond strength and water resistance on the intended substrates.
4. Approve the formulation.
5. Record the approved formulation.

10. Preparation of Material

1. Check the batch number and expiry date and record them.
2. Stir the drum or IBC before use.
3. Weigh the required quantity on a calibrated scale.

11. Use in Production

1. Charge Addmix VAE into the mixer.
2. Add plasticisers, fillers and additives slowly under stirring as per the formulation.
3. Adjust viscosity with water where required.
4. Mix until homogeneous.
5. Record the actual quantities in the batch record.

12. Batch Release

- Test the finished batch for viscosity, pH, solids and bond strength.
- Release only when results meet the specification.
- Keep a retained sample of each batch.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Material	On receipt	Milky dispersion, solids 54 – 56 %, within shelf life	QC Lab
Formulation	Each batch	As approved	Production Supervisor
Finished adhesive	Each batch	Within specification	QC Lab

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Appearance, solids, viscosity, pH	Each delivery	Goods receipt report
Weighing	Quantities	Each batch	Batch record
Finished batch	Viscosity, pH, solids	Each batch	QC report
Performance	Bond strength, open time	Each batch	QC report

14. Defects & Remedies

Defect	Likely Cause	Remedy
Coagulation / lumps	Freezing; high shear; incompatible additives	Discard affected material; check compatibility
Low bond strength	Wrong formulation	Review formulation in laboratory
Viscosity drift	Water loss or contamination	Keep containers closed

15. Health, Safety & Environment

Hazard	Control Measures
Eye and skin contact	Gloves and goggles; eye-wash available
Slippery floors	Absorb spills immediately
Manual handling	Drum trolley, forklift, correct lifting
Rotating mixers	Guards; lock-out before cleaning

- Wear gloves and safety goggles.
- Avoid contact with eyes and prolonged skin contact.

- Spills make floors slippery; absorb with inert material.
- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean equipment with water before the dispersion dries.

17. Limitations

- Do not freeze; store at +5 °C to +30 °C.
- Shelf life 9 months.
- Any change to the formulation must be approved by the laboratory.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Material received and inspected	☐ Yes ☐ No	
Formulation approved	☐ Yes ☐ No	
Container stirred	☐ Yes ☐ No	
Quantities weighed	☐ Yes ☐ No	
Additives added slowly	☐ Yes ☐ No	
Batch tested	☐ Yes ☐ No	
Batch record completed	☐ Yes ☐ No	
Retained sample kept	☐ Yes ☐ No	

Inspected by	Signature	Date

Construction Chemicals

Technical Data Sheet

Addmix VAE — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Bonding Agents



Addmix VAE

Vinyl Acetate–Ethylene (VAE) Polymer Dispersion

Product Description

Addmix VAE is a vinyl acetate–ethylene copolymer dispersion stabilised with polyvinyl alcohol. It is designed for flexible, water-resistant adhesives with rapid setting, good adhesion to plastics and a very low film-forming temperature.

Uses

- Film-to-wood lamination.
- Flooring installation adhesives.
- Paper and packaging adhesives.
- Leather, textile and foam bonding.
- Water-resistant adhesives for building insulation materials.

Advantages

- Excellent elongation, flexibility and water resistance.
- Rapid setting compared with PVAc.
- Good adhesion to plastics.
- Permanently flexible adhesive films.
- Film formation at 0 °C.

Application

1. **Formulation**
   Use as the main binder in water-based adhesive formulations.
2. **Mixing**
   Add fillers, plasticisers and additives under stirring as per the formulation.
3. **Testing**
   Verify viscosity, open time and bond strength on the intended substrates.

Consumption

As per formulation
binder in water-based adhesives
Determine the dosage by laboratory trials.

Packaging

200 kg and 220 kg drums; IBC 1,000 – 1,100 kg; flexi-bags and bulk tankers.

Storage & Shelf Life

9 months in original, unopened packaging. Store at +5 °C to +30 °C; protect from frost and direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Polymer	Vinyl acetate–ethylene copolymer	—
Protective colloid	Polyvinyl alcohol (PVOH)	—
Appearance	Milky white dispersion	—
Solids content	54 – 56 %	—
Viscosity (25 °C)	1300 – 2000 mPa·s	—
pH	4 – 6	—
Density (20 °C)	approx. 1.07 g/cm³	—
Minimum film-forming temperature	0 °C	—
Glass transition temperature	approx. 0 °C	—
Average particle size	approx. 1 µm	—
Dispersion film	Clear, glossy, slightly tacky	—

Consumption

Application	Use
Lamination adhesives	Main binder
Flooring adhesives	Main binder
Paper and packaging	Main binder
Leather / textile / foam	Main binder

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Store and use at +5 °C to +30 °C.
- Do not freeze.
- Check compatibility with all formulation components before production.
- Stir before use.

Application Notes

- Determine the formulation and dosage by laboratory trials.
- Add plasticisers and fillers slowly under stirring.
- Adjust viscosity with water where needed.
- Test bond strength and water resistance on the actual substrates.

Health & Safety

- Wear gloves and safety goggles.
- Avoid contact with eyes and prolonged skin contact.
- Spills make floors slippery; absorb with inert material.

First Aid & Cleaning

- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.
- Ingestion: rinse mouth and seek medical advice.
- Clean equipment with water before the dispersion dries.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

Addmix VAE — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

Addmix VAE — Submittal Document

To be attached: approvals and certificates, where available.